

**ADAPTIVE FRICTION COMPENSATION USING EXTENDED
KALMAN-BUCY FILTER FRICTION ESTIMATION: A COMPARATIVE STUDY**

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ABSTRACT

This paper presents an adaptive friction compensation method that does not require a structured nonlinear friction model. An extended Kalman-Bucy filter (EKF) relies on an accurate model of system rigid-body dynamics and measured motion to estimate an external friction torque. The friction estimate is used in a traditional friction compensator to cancel friction effects. The paper motivates and describes the estimation method and reports results of friction compensation experiments. In addition, the EKF compensator is compared with other friction compensation strategies, both model-based and non-model based. Experimental results for each compensator are presented for position tracking with velocity reversals. When motion is dominated by static and Stribeck friction, the EKF-based compensator consistently provides more than fifty percent RMS tracking error reduction over a well-tuned PID compensator, and equal or superior tracking performance over other compensation methods.

1. INTRODUCTION

Precision motion control requires compensation of complex friction behavior evident in a variety of systems, from machine tool positioning to pointing and tracking in airborne navigation systems. To reduce the need for high gain control laws, friction compensators typically include a friction observer that provides a friction cancellation term to be added to the control input provided by a series compensator. In recent years, there has been progress in developing adaptive friction compensation methods based on on-line estimation of parameters of dynamic friction models. As the effectiveness of the friction model improves, it is believed that the accuracy of the friction estimate improves and hence motion control error is reduced. Leonard and Krishnaprasad (1992) present a comprehensive comparative study of adaptive friction compensators of that period, including methods based on an asymmetric kinetic+viscous friction model, and the Dahl friction model (Dahl, 1977). Comparisons of different friction compensation methods are made in the context of a position tracking experiment for a DC motor. Their results show that adaptive friction compensation used with PID control provides significant rms error reduction in position tracking experiments involving repeated zero velocity crossings compared to an optimized PID compensator alone, with or without dither. Since that study, Canudas de Wit et al. (1995) proposed the LuGre model – a six-parameter dynamic state friction model that captures the Stribeck effect, hysteresis, finite stiffness behavior exhibited in static friction, and dynamic behavior in breaking static friction. The challenge in using the LuGre model is in estimating a parameter governing Stribeck effect that enters nonlinearly, as well as in estimating *all* six parameters of the friction model simultaneously. Use of the LuGre model for friction compensation has been independently

demonstrated by several authors, including Canudas de Wit and Lischinsky (1997), Feemster et al. (1998), and Misovec and Annaswamy (1998). In each case, assumptions are required, for example constant or slowly time varying model parameters, and persistent excitation. Each method estimates one or more static friction parameters, thus two dynamic friction parameters must be known by off-line evaluation (as in Misovec and Annaswamy) and must remain constant, or dynamic effects must be neglected (as in Feemster et al.). The maximum number of friction model parameters estimated is four (Misovec and Annaswamy).

Mentzeolopoulou and Friedland (1994), and Amin et al. (1997) also present experimental results of dynamic friction compensation. The first paper compares four friction observers, all with relatively simple structures – a Coulomb friction observer, an observer that tracks friction force as if it were an unknown constant, a reset integrator model based observer, and a Dahl model observer. All are found to reduce error in experiments involving position commands, where like the present study, a friction measurement is available with which to compare friction estimates. The Coulomb friction observer is considered the best of the four observers. Amin et al. (1997) further develops the Coulomb friction observer and applies it to position and velocity tracking control experiments.

In addition to model-based adaptive friction estimation, there are non-model-based methods for estimating nonlinear friction. Here, we refer to non-model-based observers as those that do not require a structured, nonlinear model bearing some relationship to the underlying physical phenomenon. Yeh and Youcef-Toumi (1998) present an adaptive control scheme based on local function estimation. Model reference adaptive control compensates the linear portion of the plant, and a local estimate of the nonlinear external input provides cancellation compensation. Though that paper applies the method to shaft positioning using magnetic bearings, it appears to be applicable to friction compensation. Ray and Remine (1998) present a friction estimation method using extended Kalman-Bucy filtering. With both of these methods, which are reviewed below, a nonlinear friction model need not be posed, a friction estimate history is provided, and friction estimates can be used for compensation. Non-model-based methods address friction characterization in the presence of variable and uncontrollable factors, such as wear, temperature, humidity, lubricant condition, and sudden or slow normal force variation. Moreover, non-model-based methods accommodate multiple sources of friction or other nonlinearities that may not be captured adequately by a single structured friction model.

The objectives of this paper are twofold. First, we investigate whether friction estimation performance is, in fact, a function of friction model complexity. Second, we determine whether better friction estimates result in improved friction compensation. To do so, we compare the performance of

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nonlinear model-based and non-model-based adaptive friction estimation and compensation methods. In the spirit of the comparative study of Leonard and Krishnaprasad, we include in our study a baseline PID compensator with no friction cancellation term, and the PID compensator with cancellation terms provided by five model-based friction observers, and two non-model-based observers. While other worthy friction estimators exist, these are chosen both to bound a hierarchy of model complexity from the Dahl model to the more recent LuGre model and to provide continuity with other comparative studies.

With our experimental facility, which consists of a DC motor driven inertia, each algorithm is tested under a variety of dominant friction regimes, ranging from Coulomb dominated friction to static and Stribeck dominated friction. Friction sources include the motor and apparatus bearings plus an optional external nonconformal (line) contact. Two redundant transducers provide friction measurements at the external contact for qualitative verification of friction force estimates. These measurements serve also to document actual friction variability during experiments, which has not been known in studies by previous authors, with the exception of the studies by Mentzeolopolou and Friedland and Amin et al.

Section two describes the experimental facility and model, and section three describes each adaptive friction compensation method considered in this paper. The experimental procedure and results are given in sections four and five.

2. EXPERIMENTAL FACILITY AND MODEL

An apparatus was designed to compare model and non-model-based adaptive friction compensators. The apparatus, pictured in Figure 1, consists of a 2.28 kg solid rotating disk made of 52100 steel with a Brinell hardness of 192, ground to be concentric on its shaft to 10^{-4} in. A DC motor drives the shaft, thus bearing friction sources include the motor bearing and two shaft support bearings. An optional cylindrical rider in a line contact with the disk and under a static normal load produces an additional friction source. In this study, an aluminum rider is used. The external contact can be operated with or without lubrication. A 2500 PPR optical encoder with fourfold multiplication determines shaft angular position with a resolution of $6.3\text{e-}4$ rad. The rider is mounted to the end of an I-beam, and additional transducers include a piezoelectric strain sensor mounted to the I-beam for high frequency friction measurements, and a capacitive displacement transducer that measures the deflection of the beam at a point just above the rider. These transducers provide indirect friction measurements at the contact, which are used for *qualitative* verification of friction force estimates. The I-beam's resonant frequency is larger than any driving frequencies. Signal amplification and 16-bit A/D converters provide adequate friction force resolution.

The apparatus is intended to study friction behavior for unidirectional or bi-directional motion with friction characteristics in the boundary, mixed, and elastohydrodynamic regimes. The encoder provides measurements of disk angular position that are used for each friction observer. The friction force measured in the apparatus is used *only* for qualitative comparison to estimated friction force. The experimental apparatus is modeled as a second-order system, driven by an input voltage to the motor, V_{in} , and affected by an unknown, nonlinear external friction torque T_f :

$$\begin{bmatrix} \ddot{\theta} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -B/J & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \theta \end{bmatrix} + \begin{bmatrix} K/J \\ 0 \end{bmatrix} V_{in} - \begin{bmatrix} 1/J \\ 0 \end{bmatrix} T_f \quad (1)$$

θ , and $\dot{\theta}$ are shaft angular position and velocity, respectively. The motor time constant is determined from the system response to a step input with no friction contact. Physical constants J (moment-of-inertia of the motor and load), B (viscous damping in the bearings), and K are determined based on the disk dimensions and material, the measured time constant, and the steady-state velocity response to step voltage inputs. Measured steady-state friction torque behavior for the apparatus with the rider up and with the rider in contact are reported in section four.

3. ADAPTIVE COMPENSATION METHODS

Baseline PID Compensator (PI)

The baseline PID compensator, designated PI, was designed to minimize rms position tracking error $e(t) = \theta_d(t) - \theta(t)$ for a sinusoidal position command $\theta_d(t)$ subject to constraints on control effort (no saturation) and closed-loop bandwidth (larger than 1 Hz). It includes a feedforward acceleration term:

$$u_{pid} = K_p e + K_i \int e dt + K_d \dot{e} + c \ddot{\theta}_d \quad (2)$$

Control gains selected are $K_p = 20$, $K_i = 10$, and $K_d = 2$, which results in a closed-loop bandwidth of 3.2 Hz. The baseline controller is used in combination with each friction observer under investigation to provide a total input signal

$$V_{in} = u_{pid} + \frac{1}{K} \hat{T}_f \quad (3)$$

$\hat{T}_f$ is the friction torque estimate. Position tracking simulation of the PID controlled motor without nonlinear friction provides rms position tracking errors of $9.07\text{e-}4$, $1.56\text{e-}4$, and $7.7\text{e-}3$ rad, respectively for 20 degree peak-to-peak sinusoidal position reference trajectories at 0.1 Hz, 0.3 Hz, and 1.0 Hz, respectively.

EKBF-Based Friction Compensation (KF)

A non-model-based friction estimator is constructed using EKBF by treating friction torque as an unknown state element and estimating the augmented state. By measuring motion along with applied forces, and knowing the system parameters (J , B , and K in eq. 1, above), one can estimate the external (unknown) friction torque required to produce the measured motion. This

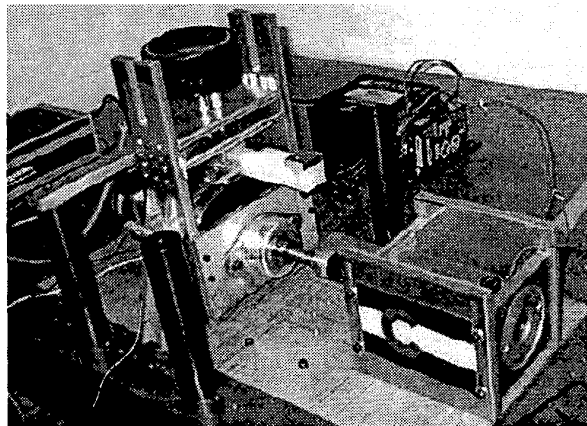


Figure 1 Experimental apparatus (support removed to show rider)

method has been proven in other applications, including estimating aerodynamic forces (Sri-Jayantha and Stengel, 1988, Stalford, 1981), and estimating tire forces in ground vehicles (Ray, 1995, 1997); however, use of the resulting estimates has been limited to modeling in the above papers. Here, the intent is to use the estimates for friction compensation. The EKBF poses a real-time estimator that optimally filters noise and estimates the friction torque along with θ and $\dot{\theta}$. The friction torque state is modeled as a random walk of order n :

$$\begin{bmatrix} \dot{T}_f \\ \ddot{T}_f \\ \vdots \\ T_f^n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} T_f \\ \dot{T}_f \\ \vdots \\ T_f^{n-1} \end{bmatrix} \quad (4)$$

$\dot{T}_f = 0$ is the simplest such random walk model, which treats friction as a random constant. Higher-order models allow the friction torque to respond adequately to transients. Eqs. 1 and 4 form the basis for a dynamic estimator for the augmented state $x = [\theta \ \dot{\theta} \ T_f \ \dot{T}_f \ \dots \ T_f^{n-1}]^T$. At each time step, the EKBF propagates the augmented state equations along with the matrix differential equation governing the error covariance; computes an optimal filter gain; and updates the state estimate and covariance matrix (Maybeck, 1982). Measurement residuals drive the random walk equations, and the process noise covariance is appended with terms for each augmented state element:

$$Q = \text{diag}[Q_\theta \ Q_{\dot{\theta}} \ Q_{T_f} \ Q_{\dot{T}_f} \ \dots \ Q_{T_f^{n-1}}] \quad (5)$$

Terms associated with the friction torque state element and its derivatives serve as tuning parameters that trade friction torque tracking ability with noise content in the estimated friction torque. The linearized state transition, control effect, and disturbance input matrices are constant for the augmented system independent of the order of the random walk model; therefore, once Q is fixed, filter gains can be precomputed, and only the state propagation and update equations need to be implemented in real-time. In all results presented in this paper, a second-order random walk model is used.

Adaptive Control using Local Function Estimation (YY)

The method of Yeh and Youcef-Toumi (1998) uses model reference adaptive control (MRAC) to make the linear part of the plant dynamics match those of a linear reference model. In addition, an adaptive cancellation signal is generated to cancel the nonlinear input. The plant and the reference model are represented by the state equations,

$$\dot{x} = Ax + f(x) + Bu \quad (6)$$

$$\dot{x}_m = A_m x_m + B_m r \quad (7)$$

The input u consists of three terms,

$$u = -Kx + \Delta r + u_{ad} \quad (8)$$

Kx is state feedback, and Δ is a feedforward term such that $A - BK = A_m$; $B\Delta = B_m$. r is the reference input. u_{ad} is an adaptive term that cancels $f(x)$, the unknown nonlinear function, based on an estimate of the first order Taylor series approximation of $f(x)$ around x_i . Error dynamics $e = x - x_m$ therefore depend on the error in estimating $f(x)$,

denoted $\hat{f}(x, x_i)$ as well as the error in approximating $f(x)$, denoted $f_e(x, x_i)$:

$$\dot{e} = (A - BK)e + f_e(x, x_i) + \hat{f}(x, x_i) \quad (9)$$

The adaptation mechanism works as follows. A certain error tolerance ρ is specified, which defines an n -dimensional tolerance sphere $\Phi(t) = \{x \in \mathbb{R}^n \mid \|x - x_m(t)\| \leq \rho\}$ centered around the reference model trajectory $x_m(t)$. When the state leaves $\Phi(t)$, the current state x_i is sampled and adaptation is switched on until the state is within $\Phi(t)$. u_{ad} is given by

$$Bu_{ad} = -\hat{f}(x, x_i) = B(\hat{C}_0(x_i, t) + \hat{C}_1(x_i, t)(x - x_i)) \quad (10)$$

The adaptation law for the coefficient vector is

$$\begin{aligned} \dot{\hat{C}}_k(x_i, t) &= -\gamma B^T P e w_k(x, x_i), \quad \|e\| \geq \rho \\ &= 0, \quad \|e\| < \rho \end{aligned} \quad (11)$$

with $w_0(x, x_i) = [1 \ \dots \ 1]^T$, $w_1(x, x_i) = x - x_i$. P is a positive definite solution to a Lyapunov equation $Q = -(PA + A^T P)$, and γ controls adaptation rate. In our implementation of the controller, the state vector includes the velocity error, the position error and the integral of the position error, thus in the absence of the adaptive input u_{ad} , the controller implements the PID compensated system. Only the position error is used to determine whether adaptation should be switched on. This corresponds to scaling of state variables to improve performance as suggested by Yeh and Youcef-Toumi (1998). In addition, we found that when tuned to minimize rms tracking error, the controller operates close to a limit cycle. A "limit cycle killer" was implemented to assure that the state did not transition to a limit cycle. The killer limits the adaptation signal when it becomes larger than an established threshold.

Dahl Friction Model (DM)

Walrath (1984) proposes a compensation method for gimbal bearing friction based on the Dahl friction model (Dahl, 1977). This method is also a candidate in the comparative study by Leonard and Krishnaprasad (1992). The Dahl model is condensed to a two-parameter, first-order model for friction torque:

$$\tau \frac{dT_f}{dt} + T_f = T_c \text{sgn}(\dot{\theta}) \quad (12)$$

$$\tau = \frac{T_c}{\sigma |\dot{\theta}|} \quad (13)$$

τ depends on the rest stiffness $\sigma = \frac{dT_f}{d\theta} \Big|_{\dot{\theta}=0}$, the Coulomb

friction T_c , and velocity. To capture velocity dependence, both Leonard and Walrath find values of τ that minimize rms position tracking error as functions of rms velocity and acceleration, respectively. The value of τ is then updated based on the reference input trajectory. T_c must be identified and the empirical relationship between τ and rms velocity or acceleration must be derived for *each loading condition*, thus adaptation in this compensator is limited to the empirical relation between model parameters and velocity. Nevertheless, it is included here to provide some overlap with the previous comparative study of Leonard and Krishnaprasad, and because when well-tuned, performance is quite good.

Coulomb Friction Observer (CF)

The nonlinear observer of Amin et al. (1997) takes the form

$$\begin{aligned}\hat{T}_f &= \hat{a} \operatorname{sgn}(\dot{\theta}) \\ \hat{a} &= z + K_f |\dot{\theta}| \\ \dot{z} &= K_f (u - \hat{T}_f) \operatorname{sgn}(\dot{\theta})\end{aligned}\quad (14-16)$$

u is the total control torque input to the motor, and K_f is a selectable gain. In the absence of a velocity measurement, a velocity observer based on position measurements is proposed (Tafazoli et al., 1995):

$$\frac{\dot{\theta}(s)}{\theta(s)} = \frac{k_v s}{s + k_v} \quad (17)$$

Since all compensation methods except the KF and YY require velocity as an input, the high-pass filter of eq. 17 is used to provide velocities from measured position. The CF observer is one of the simplest friction observers to implement, as it requires the selection of only two decoupled gains, K_f and k_v .

LuGre Friction Model (LS, LD, and LA)

Canudas de Wit and Lischinsky (1997), Feemster et al. (1998), and Misovec and Annaswamy (1998) implement adaptive friction compensators that estimate parameters in the LuGre friction model (Canudas de Wit et al. 1995):

$$\begin{aligned}T_f &= \sigma_0 z + \sigma_1 \dot{z} + \alpha_2 \dot{\theta} \\ \dot{z} &= \dot{\theta} - \sigma_0 \frac{z |\dot{\theta}|}{g(\dot{\theta})} \\ g(\dot{\theta}) &= \alpha_0 + \alpha_1 e^{-(\dot{\theta}/\dot{\theta}_s)^2}\end{aligned}\quad (18-20)$$

σ_0 and σ_1 represent stiffness and viscous damping coefficients, respectively, for asperity deflection during presliding displacement. α_0 is the Coulomb friction, $\alpha_0 + \alpha_1$ is the static friction, α_2 is the viscous friction coefficient, and $\dot{\theta}_s$ is the Stribeck velocity. Three friction compensators based on the LuGre model are included here. The first, denoted LuGre Static (LS), neglects dynamic friction (i.e., assumes $\dot{z}=0$) and uses static friction parameters ($\alpha_0, \alpha_1, \alpha_2$, and $\dot{\theta}_s$) measured under steady-state conditions to estimate T_f as a function of velocity. Parameters are not adapted, but separate static parameter estimates are used for the contact up and contact down cases. The second compensator, LuGre Dynamic (LD), retains the dynamic term, and again, parameters are not estimated on-line. The final compensator, LuGre Adaptive (LA), uses the adaptation method proposed in Misovec and Annaswamy (1998) to estimate α_0, α_1 and α_2 . The parameter adaptation law is

$$\dot{\Theta} = \Gamma W(\dot{\theta}) B^T P e \quad (21)$$

where

$$\begin{aligned}\Theta &= [\alpha_0 \quad \alpha_1 \quad \alpha_2]^T \\ W &= \begin{bmatrix} \operatorname{sgn}(\dot{\theta}) & \operatorname{sgn}(\dot{\theta}) e^{-(\dot{\theta}/\dot{\theta}_s)^2} & \dot{\theta} \end{bmatrix}\end{aligned}\quad (22-23)$$

$e = \left[\int e dt \quad e \right]^T$, B is the control effect matrix for the error dynamics, which are represented by $\{A, B\}$, and P is the positive definite solution to the Lyapunov matrix equation

$Q = -(PA + A^T P)$. The asperity stiffness and damping parameters must be known a priori. The experimentally measured parameter $\dot{\theta}_s$ is assumed to be constant, as is the inertia J . The limit cycle killer described above was implemented along with this compensator, as a tendency towards unstable behavior resulted when parameter estimates drifted due to less than adequate excitation. By including three versions of the LuGre compensator, we aim to determine the performance of model-based friction estimation based on a good physical model of friction with good parameter estimates; the role of the dynamic terms in friction estimation; and changes in performance due to parameter adaptation. In essence, the LuGre Dynamic model represents the best nonlinear dynamic friction model currently available for a system with stationary parameters.

4. EXPERIMENTAL PROCEDURE

Performance of each controller is examined for sinusoidal reference input $\theta_d(t) = A \sin \omega t$, with amplitude $A = 0.1745$ rad (10 degrees) and three different frequencies: 0.1 Hz, 0.3 Hz, and 1.0 Hz. Performance is examined both with and without the external rider contact. The lowest frequency input provides stick-slip dominated motion in which static friction and Stribeck effect are pronounced with the contact down. With the contact up, friction behavior is generally dominated by Coulomb and viscous friction, especially at higher frequency inputs. For the LuGre compensators, estimates of α_0, α_1 , and α_2 , and an estimate of the Stribeck velocity $\dot{\theta}_s$ are required. To estimate these parameters, as well as to characterize the apparatus, constant velocity control experiments were performed at a range of velocities from 0.15 rad/sec to 12 rad/sec, following the method outlined in Canudas de Wit and Lischinsky (1997). The input torque vs. velocity setpoint data were used to optimize static parameters of the LuGre model to minimize the squared error between the model predicted friction torque and the data points. Resulting parameter estimates with and without the external contact are given in Table 1. Figure 2 shows data points and the fitted curves. These off-line identified static parameters are used in all three LuGre compensators. For the LA compensator, the parameters serve as initial conditions for parameter adaptation. The LD and LA compensators require estimates of the dynamic parameters σ_0 and σ_1 in addition to the static parameters. These were chosen to minimize rms tracking error for the 0.3 Hz trajectory. Separate parameter values were estimated for the contact up and contact down cases.

The Dahl friction observer (DM) requires parameter T_c , and the relationship between τ of eq. 12 and the reference input velocity. T_c is equivalent to α_0 , and the value of τ that minimizes RMS tracking error when used in combination with the PID control law is determined for each reference input frequency and each contact condition. Walrath (1984) found the optimum value of $1/\tau$ to be linearly related to rms acceleration, while Leonard and Krishnaprasad (1992) found $1/\tau$ to be linear with rms velocity. We observed a broad minimum in rms tracking error vs. $1/\tau$ for each input trajectory, and the velocity relationship was difficult to establish for the contact up condition. The optimizing parameters are given in Table 2. For estimating τ , rms error calculations are based on a minimum of four and a maximum of ten complete cycles.

All other constant gains or tuning parameters are chosen to minimize rms tracking error for a 0.3 Hz sinusoidal position trajectory, with the contact up. The constants are *not changed* for the remaining frequencies with the exception of the non-adaptive methods—DM, LS, and LD. Table 1 lists the remaining parameters for each compensator. Table 1 is presented in increasing order of our subjective rating of complexity, where complexity relates to the number of parameters and the difficulty of selecting these parameters in order to obtain adequate closed-loop performance and stability. A 0.01 sec sample time is used throughout the experiment. All compensators are implemented using a dSPACE DS1102 DSP board.

5. EXPERIMENTAL RESULTS

Table 2 and Figure 3 summarize tracking results for each compensator, input frequency, and contact condition. Mean and standard deviation of rms position tracking error are based on five samples taken over the course of two days. Many more samples were taken over several months in the process of implementing and tuning compensators; however, once tuning was complete, the five samples for which statistics are provided were taken. Figure 3 reports the mean rms tracking error of Table 2 with 95% confidence intervals. RMS error for individual samples is based on four, six, and ten cycles, respectively, for the 0.1 Hz, 0.3 Hz, and 1.0 Hz inputs. Because the friction behavior displayed some variability, particularly with the contact down, rather than running five consecutive samples at a given input frequency for the same compensator, one sample was taken for *each* compensator, input frequency, and contact condition before proceeding to the next sample. Longer consecutive samples (40, 70, and 100 cycles, respectively for the three input frequencies) were also collected. The standard deviations for consecutive runs (not reported) are lower than those reported in Table 2, though mean values are comparable, thus worst case confidence intervals on mean rms tracking error are provided for nonconsecutive data.

The results show that all of the friction compensation methods provide reduction in rms position tracking error over the well-tuned PID compensator. Starting with the contact up and higher frequencies (0.3 Hz and 1.0 Hz), the DM compensator provides remarkable reduction in rms tracking error. This outstanding performance is attributed to the fact that it is largely a Coulomb friction observer, and the friction behavior for these trajectories is Coulomb friction dominated. Moreover, off-line tuning of the DM compensator parameters provides adequate

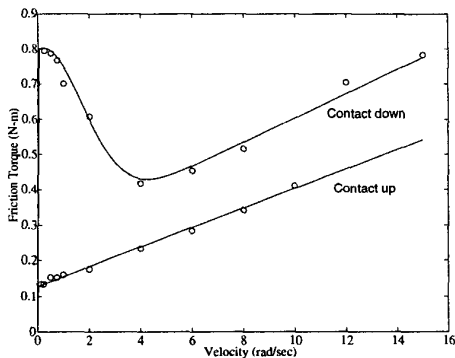


Figure 2 Steady-state friction characteristics of experimental apparatus with and without the external friction contact.

Table 1 Experimentally determined parameters for each adaptive friction estimation algorithm

Method	Parameters
CF	$K_f=0.15$, $K_v=20$
KF	$Q_{\dot{\theta}}=Q_{\theta}=0$, $Q_{T_f}=Q_{\dot{T}_f}=2e-4$
LS	<u>Contact up</u> $\alpha_0=0.128$ N-m $\alpha_1=0.088$ N-m $\alpha_2=0.0276$ N/rad/s $\dot{\theta}_s=0.01$ rad/s <u>Contact down</u> $\alpha_0=0.26$ N-m $\alpha_1=0.54$ N-m $\alpha_2=0.0344$ N/rad/s $\dot{\theta}_s=2.4$ rad/sec.
DM	<u>Contact up</u> $\tau=1/24$ s (0.1 Hz) $\tau=1/16$ s (0.3 Hz) $\tau=1/40$ s (1.0 Hz) $T_c=0.128$ N-m. <u>Contact down</u> $\tau=1/10$ s (0.1 Hz) $\tau=1/20$ s (0.3 Hz) $\tau=1/36$ s (1.0 Hz) $T_c=0.26$ N-m.
YY	γ, Q, ρ
LD	$\alpha_0, \alpha_1, \alpha_2, \dot{\theta}_s$ (same as LS) <u>Contact up</u> $\sigma_0=5$ N-m/rad $\sigma_1=0.01$ N-m/rad/s <u>Contact down</u> $\sigma_0=10$ N-m/rad $\sigma_1=0.1$ N-m/rad/s
LA	$\alpha_0, \alpha_1, \alpha_2, \dot{\theta}_s, \sigma_0$, and σ_1 (same as LD), Γ, Q

friction estimates. At 0.1 Hz contact up or with the contact down, friction behavior is no longer Coulomb dominated, and the DM is no longer the best observer.

The LS and LD compensators, which also have off-line tuned friction parameters, exhibit good performance overall. With the contact up, no statistically significant difference in the performance of these compensators is noted. With the contact down, the *static* LuGre compensator outperforms the dynamic compensator. Here, it is possible that the dynamic parameters σ_0 and σ_1 are *not* constant and thus affect dynamic friction estimates. Since the LA compensator is initialized well, it should perform as well as the LD compensator, as it does, with the exception of the 0.1 Hz, contact down input. It was noted during experimentation that the low frequency input provides the least amount of excitation, and parameter drift is significant. This sometimes results in unstable behavior and activation of the limit cycle killer described earlier.

The CF observer performs well for all cases except the 1.0 Hz no contact case, where it apparently overcompensated for friction, resulting in an increase in rms error over the PID system. The YY observer provides the least rms error reduction of all compensators tested. This is attributed largely to the limit cycle killer, which is required to maintain stability over the variety of operating conditions. For all adaptive compensators (YY, CF, LA, and KF), changing constant tuning parameters and gains for each operating condition (frequency and contact) *can* provide improved tracking; however, our objective for adaptive compensation methods was to determine performance for operating conditions that may not be known a priori.

Finally, performance of the EKF-based compensator was excellent overall, providing rms error reductions of 38% to 59% over the PID system. The EKF compensator matches or outperforms the off-line tuned, model-based LS and LD compensators in five of the six conditions tested, as seen visually in Figure 3. This is remarkable, given the fact that a nonlinear dynamic model is not required for EKF-based friction compensation. In addition, the EKF compensator generally

Table 2 Position Tracking Results

freq. (Hz)	Method	No external contact			With external contact, 5 lbf. normal load		
		Mean rms error (rad)*	Std. Dev. of rms error (rad)*	Max error (rad)**	Mean rms error (rad)*	Std. Dev. of rms error (rad)*	Max error (rad)**
0.1	PI	0.0066	1.92e-4	0.0168	0.0471	5.4e-3	0.1090
	DM	0.0041	4.57e-4	0.0131	0.0290	2.0e-3	0.0902
	CF	0.0045	5.65e-4	0.0146	0.0396	4.6e-3	0.1118
	LS	0.0029	3.05e-4	0.0100	0.0210	2.0e-3	0.0648
	LD	0.0026	4.04e-4	0.0088	0.0297	2.3e-3	0.1063
	LA	0.0028	2.31e-4	0.0156	0.0412	5.6e-3	0.1227
	KF	0.0027	1.95e-4	0.0073	0.0194	2.3e-4	0.0545
	YY	0.0062	3.64e-4	0.0131	0.0415	4.6e-3	0.1172
0.3	PI	0.0105	3.82e-4	0.0194	0.0546	8.0e-3	0.1163
	DM	0.0030	2.90e-4	0.0090	0.0339	4.2e-3	0.0749
	CF	0.0047	3.56e-4	0.0131	0.0488	6.6e-3	0.1061
	LS	0.0053	3.43e-4	0.0126	0.0255	5.2e-3	0.0676
	LD	0.0053	2.18e-4	0.0112	0.0403	2.9e-3	0.1164
	LA	0.0053	3.77e-4	0.0117	0.0467	1.02e-2	0.1140
	KF	0.0044	2.07e-4	0.0095	0.0230	1.8e-3	0.0472
	YY	0.0070	2.61e-4	0.0123	0.0343	7.2e-3	0.1013
1.0	PI	0.0118	4.74e-4	0.0196	0.0445	5.0e-3	0.0733
	DM	0.0031	4.30e-4	0.0109	0.0262	3.6e-3	0.0658
	CF	0.0332	2.6e-3	0.0558	0.0255	3.3e-3	0.0571
	LS	0.0055	4.00e-4	0.0133	0.0179	4.0e-3	0.0592
	LD	0.0056	3.29e-4	0.0124	0.0350	6.0e-3	0.0917
	LA	0.0059	4.70e-4	0.0122	0.0329	2.3e-3	0.0824
	KF	0.0073	2.00e-4	0.0127	0.0192	3.3e-3	0.0449
	YY	0.0090	3.82e-4	0.0163	0.0232	4.5e-3	0.0520

* based on 5 trials, **max of 5 trials

provides very consistent performance, as indicated by the tight confidence intervals.

Performance is further investigated by comparing total friction estimates from some compensators to contact friction measurements, as well as by examining the spectral content of the tracking error. Figure 4 shows the estimated friction torque for the 0.3 Hz input with the contact up, and four of the observers. The friction estimate of the DM observer, which outperforms all other observers for this input frequency and contact condition, shows a fast transition to Coulomb friction, for which the observer is best suited. The LS observer shows the effect of retaining the static friction term through spikes in the friction estimate at zero velocity. The LD friction estimates also show the Coulomb friction domination. The KF friction estimate contains some high frequency content, which filters into the tracking error. Note that the spectral content of the rms tracking error of the DM observer shows a less pronounced 0.3 Hz sinusoidal component, (i.e., a tendency towards white or colored noise). This suggests that the friction estimate is, in fact, accurate, though for the contact up case, no measurement is available to confirm this. The magnitude of the friction estimate is similar for all four observers.

Figure 5 shows the friction estimates for the same observers and the 1.0 Hz. contact down case. Here, the contact deflection, which is proportional to contact friction torque, is available, providing a measurement with which to compare friction estimates. (Note that the friction observer provides total friction torque, while the measurement provides friction torque at the contact only.) For the DM, LS, and LD observers, the contact friction measurement shows a pronounced, asymmetric Stribeck effect, which is not apparent in the model-based estimates. The DM compensator, which due to the assumed model, is incapable of tracking Stribeck friction, does not perform as well for this input and contact condition. The LS and KF track friction quite

well, though again, the KF introduces a high frequency component, which in turn affects the actual friction force. Both the LS and LD observers show a less pronounced Stribeck effect than that of the measurement. The tracking error of all four observers retains a dominant 1.0 Hz component, suggesting that additional performance gains could result if friction estimates improved.

We observed significantly different stability characteristics for each observer, with the LD and LA observers being the most sensitive to initial conditions or manual perturbations of the shaft during operation. Though additional study is warranted, we attribute this to the possibility that if the friction model structure or parameters are in error, it is possible to overcompensate for friction and induce instability.

CONCLUSIONS

Four conclusions are drawn from this study. First, friction compensation performance is not necessarily a function of friction model complexity. EKBF friction estimates based on the linear random walk model driven by measurement residuals compare well to measured contact displacement and to model-based estimates of friction torque. Moreover, the EKBF-based compensator performs as well as model-based compensators under five of six conditions tested. Second, better friction estimates do result in improved friction compensation, as indicated by the random nature of the tracking error for the Dahl model compensator operating when Coulomb friction is dominant. Third, the act of friction compensation itself appears

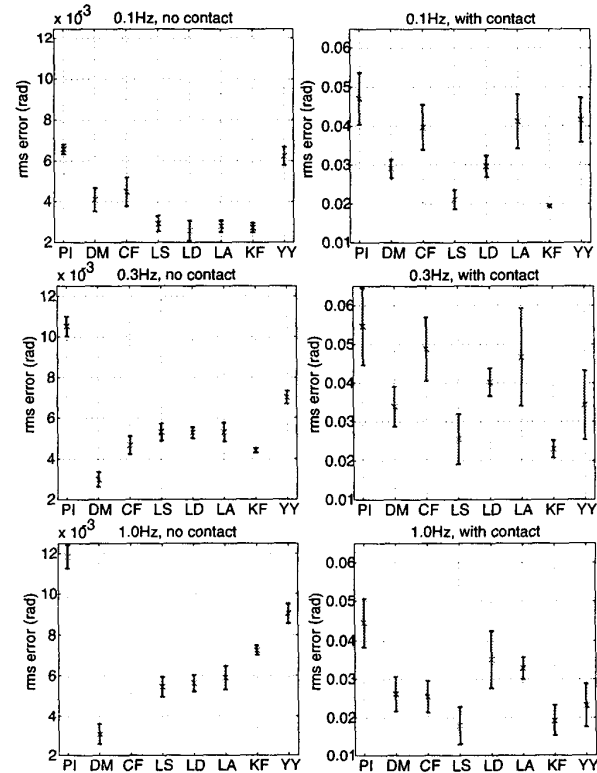


Figure 3 Mean rms tracking error and 95% confidence intervals for eight friction compensators.

to affect underlying friction characteristics. Finally, if a model-based compensator contains parameter or structure uncertainty, as is likely the case for the dynamic parameters of the LD and LA compensators, the friction estimates and resulting friction compensation performance is reduced.

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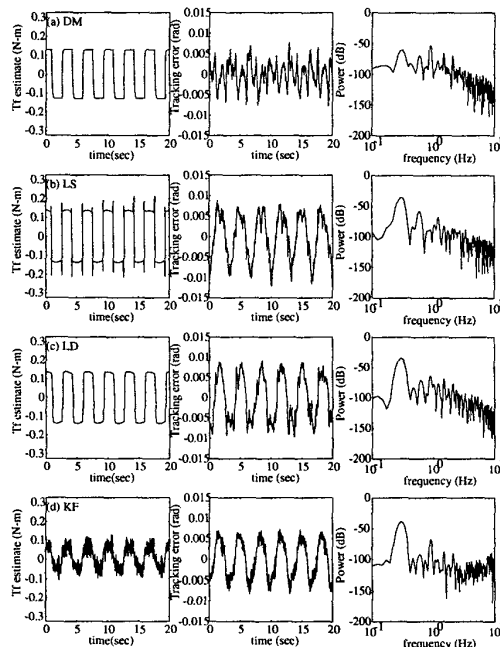


Figure 4 Estimated friction torque, tracking error, and power spectral density of tracking error for 0.3 Hz input, contact up and controllers (a) DM, (b) LS, (c) LD, and (d) KF.

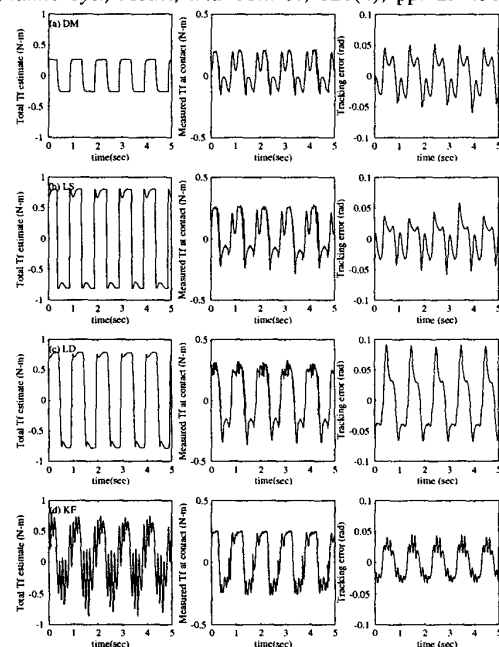


Figure 5 Estimated friction torque, measured friction torque at line contact, and tracking error for 1.0 Hz input, contact down and controllers (a) DM, (b) LS, (c) LD, and (d) KF.